

# Joshua Alvarez

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## Professional Profile

Database and backend engineer specializing in data modeling, structured storage architecture, and query optimization. Proven track record of managing enterprise relational databases, user authentication data, and high-availability operational environments.

## Technical Skills

|                |                                                                                |
|----------------|--------------------------------------------------------------------------------|
| Databases      | PostgreSQL, MySQL, H2, Relational Schema Design                                |
| Languages      | SQL, Python, Java, JavaScript, Bash                                            |
| Backend & ORMs | Hibernate, Flask, Node.js, RESTful API Integration                             |
| Data Ops       | Data Modeling, Query Optimization, User Authentication Data Handling, Azure AD |

## Professional Experience

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| May 2025–Present  | <b>IT Manager, <i>The Reeds at Shelter Haven</i>, Stone Harbor, NJ</b> <ul style="list-style-type: none"><li>○ Maintain and optimize data-driven operations including enterprise POS systems and access control databases.</li><li>○ Architect and deploy structured user accounts and permission datasets through Azure Active Directory.</li><li>○ Analyze data flow across departments to identify technical inefficiencies and streamline backend data reporting workflows.</li></ul> |
| Jan 2026–May 2026 | <b>Systems Engineer (Contract), <i>Cloud Speak</i></b> <ul style="list-style-type: none"><li>○ Designed structured backend logic for a cloud-based hospitality content management platform.</li><li>○ Engineered physical access control server infrastructure, ensuring reliable data synchronization across 3 hotel properties.</li><li>○ Maintained and troubleshoot critical system state data to preserve operational continuity.</li></ul>                                          |

## Database & Backend Projects

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| 2023–2024 | <b>Express Auto Web Application, <i>Full-Stack Integration</i></b> <ul style="list-style-type: none"><li>○ Architected and optimized a robust PostgreSQL database schema to handle application state and inventory.</li><li>○ Implemented secure user authentication handling, hashing, and session management within the relational database.</li><li>○ Integrated backend database triggers with third-party APIs (Twilio, SendGrid) for automated transactional messaging.</li></ul> |
| 2024–2025 | <b>Self-Paced AI Learning Platform, <i>REU Grant Recipient</i></b> <ul style="list-style-type: none"><li>○ Designed relational models to structure dynamic subject generation, quiz data, and student metrics.</li><li>○ Integrated user tracking and progress metrics, ensuring consistent read/write performance during automated evaluations.</li><li>○ Built instructor-style dashboards that execute complex queries to analyze aggregate student progression data.</li></ul>      |

## Education

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| Dec 2025 | <b>Bachelor of Science in Computer Science, <i>Stockton University</i>, Galloway, NJ, GPA: 3.86</b> <ul style="list-style-type: none"><li>○ <b>Honors:</b> Research Experience for Undergraduates (REU) Recipient for AI Research.</li><li>○ <b>Relevant Coursework:</b> Data Structures &amp; Algorithms I &amp; II, Software Engineering.</li></ul> |
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